

Simulation, Fabrication, and Characterization of Silicon Bragg-Grating Optical Filters

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Abstract—This proposal investigates sidewall Bragg-grating filters implemented on a 220-nm silicon-on-insulator (SOI) photonics platform. Six simulated designs are considered and grouped into two categories: cascaded Bragg gratings, and three single-grating reference. The simulations examine how the grating period and corrugation width influence the spectral response. The proposed work aims to connect nominal simulations with optical measurement results.

I. INTRODUCTION

A Bragg grating contains a periodic effective-index perturbation. Near the Bragg wavelength, reflections from successive periods add constructively and create a transmission stop band [1]. Integrated silicon implementations permit the grating period, corrugation width, number of periods, and local phase to be lithographically controlled [2], [3]. These parameters determine the center wavelength, coupling strength, spectral width, and rejection of the filter.

The central methodology of this proposal is the comparison

nominal simulation $\longrightarrow$ fabricated geometry
 $\longrightarrow$ measurement.

II. DESIGN OBJECTIVES

The nominal device consists of a 220-nm-thick silicon waveguide with a 500-nm-wide access waveguide, designed to operate with the fundamental quasi-TE mode around $\lambda_0 = 1550$ nm. The Bragg-grating study includes three main device groups: cascaded gratings, a single-grating reference, and phase-shifted Bragg gratings. The simulations are used to investigate how the grating period shifts the Bragg wavelength and how the corrugation width influences the stop-band bandwidth and extinction. Particular attention is given to narrow-band application with single etch fabrication, where the minimum practical corrugation width is investigated as a means of reducing the grating coupling strength and obtaining the narrowest achievable stop band within the fabrication and design-kit constraints. The study also examines how individual Bragg sections behave when several gratings are cascaded, especially whether long cascaded structures modify the spectral response of each individual grating through insertion loss, spectral overlap, inter-section interference, accumulated phase, or fabrication variations. After fabrication, the measured

spectra will be compared with both the nominal simulations and simulations updated using the fabricated dimensions.

III. WAVEGUIDE MODEL

The nominal waveguide mode was simulated with Lumerical MODE. The fully etched silicon waveguide has width $w = 500$ nm and height $h = 220$ nm. The solver was used to calculate the fundamental quasi-TE mode and its effective and group indices as functions of wavelength.

For a guided mode with effective index $n_{\text{eff}}(\lambda)$,

$$\beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}, \quad \phi(\lambda) = \beta(\lambda)L. \quad (1)$$

The group index is

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (2)$$

The effective index determines phase accumulation, whereas the group index determines the wavelength spacing of MZI fringes and affects the wavelength sensitivity of the Bragg condition [4], [5].

The nominal simulation gives $n_{\text{eff}}(1.55 \mu\text{m}) \approx 2.443919$ and $n_g \approx 4.195$. Figure 2 also shows the expected decrease of effective index and increase of group index over the simulated wavelength range.

The wavelength dependence of the effective index was approximated by a second-order polynomial centered at $\lambda_0 = 1.55 \mu\text{m}$:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2, \quad (3)$$

with the fitted form

$$n_{\text{eff}}(\lambda) = 2.44392 - 1.12967(\lambda - 1.55) - 0.0410869(\lambda - 1.55)^2, \quad (4)$$

where λ is expressed in micrometres.

Fabrication-corner simulations were performed for widths from 470 to 510 nm and heights from 215 to 223 nm. As shown in Fig. 1, both indices depend on the fabricated cross-section; waveguide width is especially influential over the simulated range.

IV. BRAGG-GRATING THEORY AND DESIGN

A. Bragg condition and coupling

A sidewall Bragg grating periodically changes the waveguide width and therefore the modal effective index. For a first-order grating, the approximate Bragg condition is

$$\lambda_B \approx 2n_{\text{eff}}\Lambda, \quad (5)$$

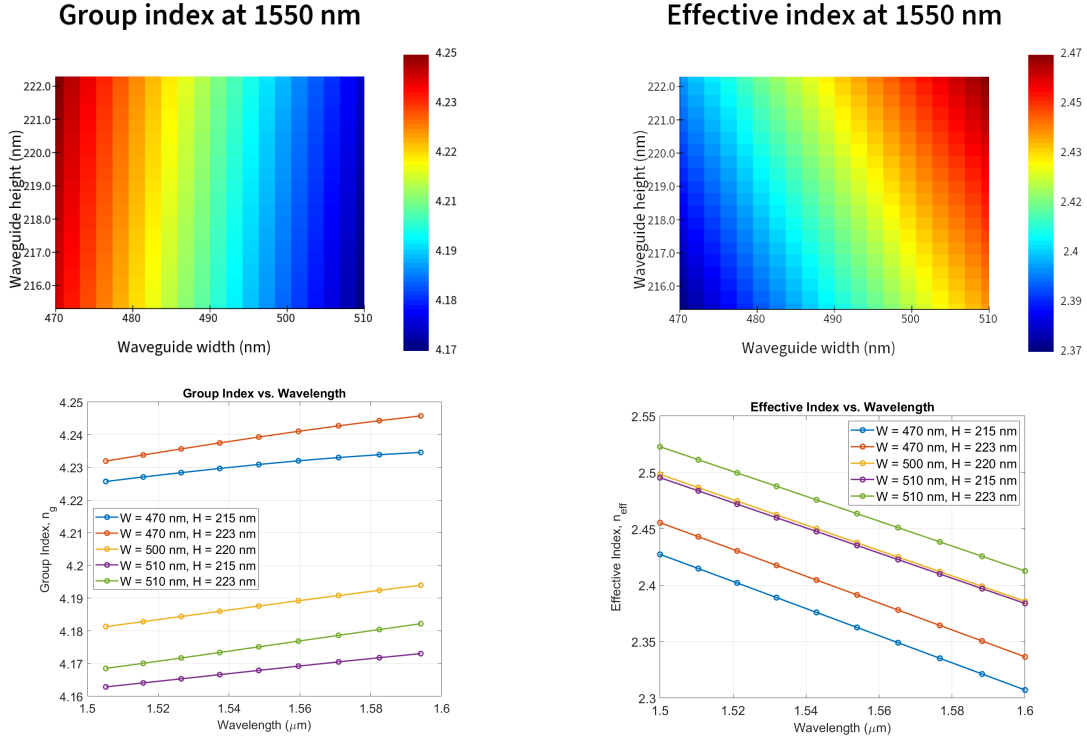


Fig. 1. Simulated dependence of group index and effective index on waveguide width, height, and wavelength. The upper panels give the indices at 1550 nm; the lower panels show wavelength sweeps for the nominal geometry and selected fabrication corners.

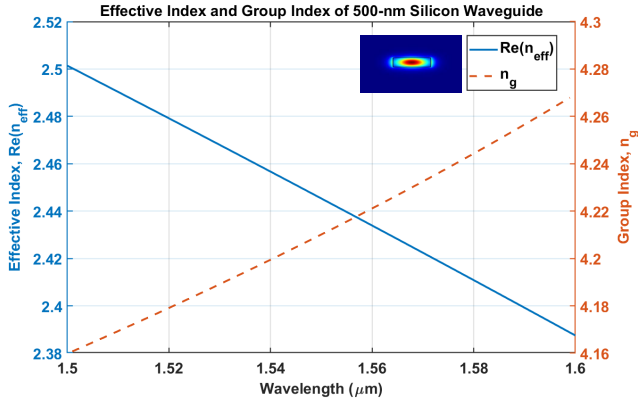


Fig. 2. Lumerical MODE result for the nominal 500 nm × 220 nm silicon waveguide. The plot shows the real part of the effective index, the group index, and the fundamental quasi-TE mode profile.

where Λ is the grating period and n_{eff} is the appropriate average or Bloch-mode effective index of the periodic section. The access-waveguide index gives only an initial estimate. With $n_{\text{eff}} = 2.443919$, a 1550-nm target gives $\Lambda \approx 317$ nm.

Equation (5) shows why changing the grating period translates the stop band. Cascading sections with slightly different periods can therefore create multiple closely spaced spectral features. If the individual stop bands are narrower than their wavelength separation, distinct rejection bands are obtained; stronger coupling or finer period spacing can instead produce overlap and a broadened or structured composite response.

The approximate bandwidth between the first nulls is [4]

$$\Delta\lambda = \frac{\lambda_B^2}{\pi n_g} \sqrt{\kappa^2 + \left(\frac{\pi}{L}\right)^2}, \quad (6)$$

where κ is the coupling coefficient and L is the grating length. For $\kappa \gg \pi/L$,

$$\Delta\lambda \approx \frac{\lambda_B^2 \kappa}{\pi n_g}. \quad (7)$$

For a sinusoidal index perturbation [4],

$$\kappa_{\text{sinusoidal}} \approx \frac{\pi \Delta n}{2\lambda_B}, \quad (8)$$

where Δn is the effective-index modulation amplitude. Increasing corrugation width generally increases Δn and κ , and can therefore increase reflectivity and broaden the stop band. Figure 3 gives the fitted design relation used to select the corrugation-width range.

B. Provisional device set

Table I separates the devices into the six experimental groups. The period lists and period counts reproduce the current simulation definitions.

V. SIMULATED BRAGG-GRATING RESPONSE

The six available plots show wavelength-dependent through responses over approximately 1500–1600 nm. The source exports use a decibel vertical scale but do not state its quantity or normalization.

TABLE I
PROVISIONAL BRAGG-GRATING DESIGNS AND THEIR ROLES IN THE STUDY

Design ID	Grating period(s) (nm)	Number of periods	Corrugation width (nm)	Design purpose
Bragg_10_4B	310, 315, 320, 325	1000 per section	10	Four cascaded, relatively narrow stop bands; period-dependence study.
Bragg_20_4B	310, 315, 320, 325	1000 per section	20	Matched four-section sweep for testing the effect of stronger corrugation on bandwidth and rejection.
Bragg_10_7B	310, 312.5, 315, 317.7, 320, 322.5, 325	1000 per section	10	Denser seven-section sweep for studying closely spaced stop bands and spectral overlap.
Bragg_20_230_500	320	500	20	Single-grating fabrication and measurement reference; baseline for the cascaded devices.
Bragg_20_230_1000	320	1000	20	Single-grating fabrication and measurement reference; baseline for the cascaded devices.
Bragg_20_315_1000	315	1000	20	Single-grating fabrication and measurement reference; baseline for the cascaded devices.

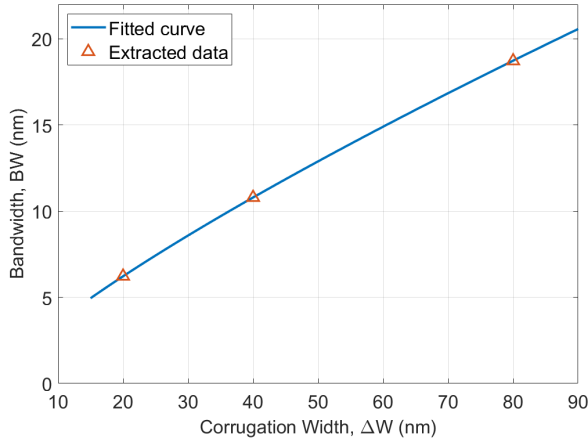


Fig. 3. Estimated Bragg stop-band width as a function of corrugation width for an oxide-clad $500 \text{ nm} \times 220 \text{ nm}$ strip waveguide. Extracted data are compared with the fitted electron-beam-lithography design model [4].

A. Cascaded Bragg gratings

Figure 4 shows four distinct stop bands for Bragg_10_4B. Their minima occur at approximately 1529, 1543, 1558, and 1572 nm, and the plotted response recovers between adjacent bands. Thus, for the simulated 10-nm corrugation and the selected periods, the cascade produces separated narrow-band features rather than one continuous rejection band. The minima approach or pass -60 dB on the plotted scale.

The corresponding 20-nm corrugation design is shown in Fig. 5. The four stop bands occur at comparable wavelengths, as expected because the grating periods are unchanged. Relative to Fig. 4, each rejected region is visibly wider and the minima reach the -80 dB lower limit of the plotted scale. The simulation therefore supports, for this matched pair, the expected increase in coupling strength, bandwidth, and rejection with corrugation width. The bands remain distinguishable, but their wider edges reduce the pass regions between adjacent gratings.

The denser period sweep in Fig. 6 produces seven discernible stop bands between approximately 1528 and 1573 nm. The features remain individually identifiable, but the recovery intervals are smaller and the composite response is more structured than for the four-section device. This design is

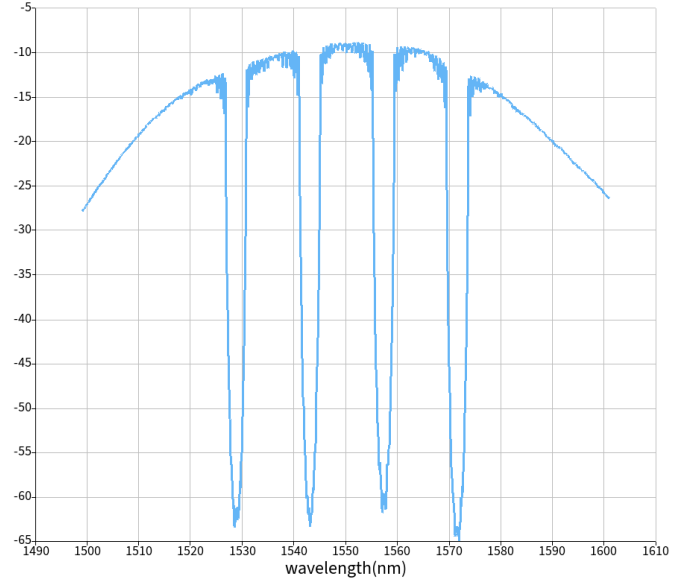


Fig. 4. Simulated through response of Bragg_10_4B: four cascaded Bragg sections with 10-nm corrugation width and four grating periods. Four separated stop bands translate across the wavelength range in accordance with Eq. (5).

therefore useful for determining experimentally when period-spaced Bragg resonances begin to interact or overlap.

B. Single-grating reference

Two single Bragg gratings, each consisting of 1000 periods with grating periods of $\Lambda = 315 \text{ nm}$ and $\Lambda = 320 \text{ nm}$, are also included as reference devices. These structures provide a baseline for comparing the spectral response of the individual gratings with that of the cascaded configuration and help identify the effects introduced by cascading multiple Bragg sections. The 500-period reference in Fig. 7 has one stop band centered near 1557 nm, with a plotted minimum of approximately -52 dB .

VI. PROPOSED LAYOUT AND FABRICATION

VII. OPTICAL CHARACTERIZATION

VIII. COMPARISON AND ANALYSIS PLAN

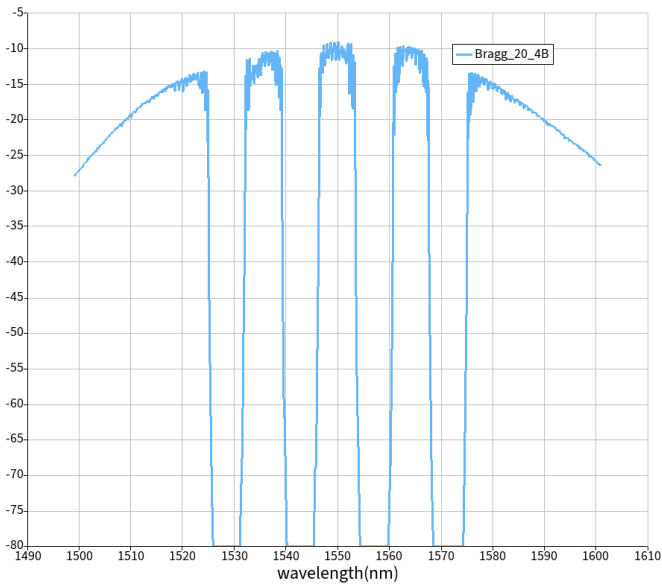


Fig. 5. Simulated through response of Bragg_20_4B. Increasing the corrugation width from 10 to 20 nm broadens and deepens the four rejection bands while retaining their period-dependent center wavelengths.

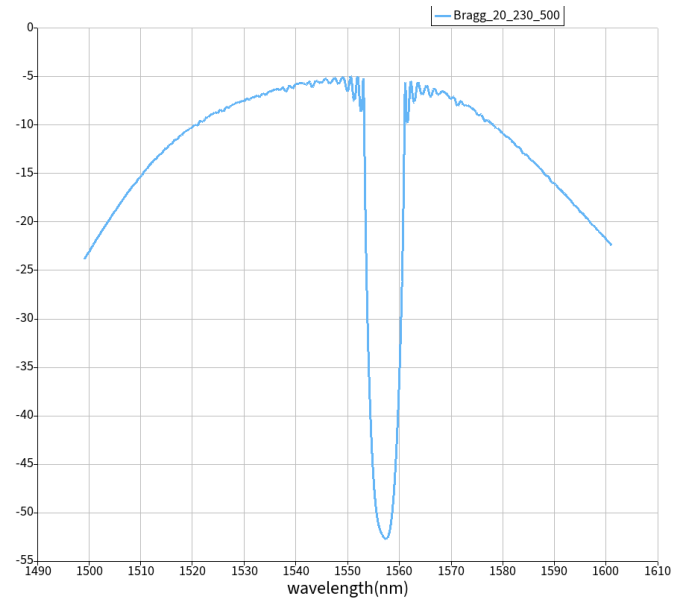


Fig. 7. Simulated through response of the single 500-period Bragg_20_320_500 reference with 20-nm corrugation width and a nominal period near 320 nm.

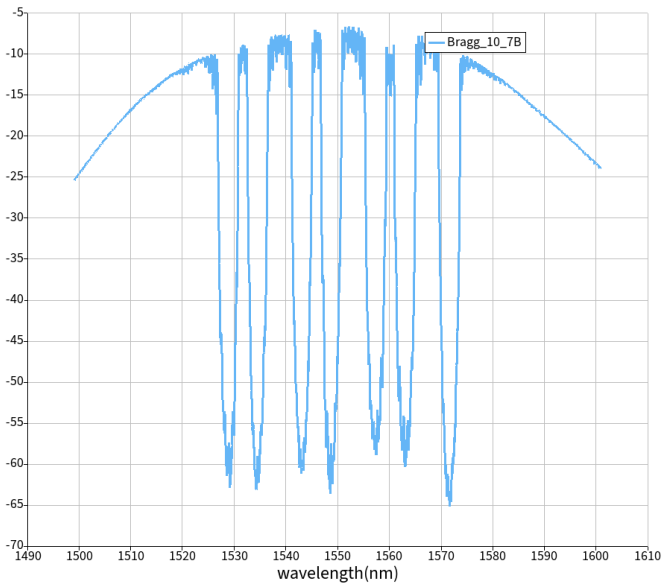


Fig. 6. Simulated through response of Bragg_10_7B. Seven more closely spaced period values produce a denser set of distinct rejection features and a structured composite spectrum.

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IX. CONCLUSION

DISCLOSURES

DATA AVAILABILITY

REFERENCES

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